

GRA-Hormonal Explosion of Subject: A Generative Regulatory Automaton for Simulating Endocrine Burst Dynamics

qqewq
Independent Researcher
GitHub: qqewq
Project Repository

Abstract—Biological systems exhibit complex feedback-driven hormonal surges that govern critical physiological and behavioural transitions. This paper introduces the GRA-Hormonal Explosion of Subject (GRA-HES) framework, a generative regulatory automaton designed to model, visualize, and analyse acute endocrine outbursts in a virtual subject. The model combines a genetic regulatory algorithm (GRA) with a layered hormone–receptor interaction graph, enabling cascading, self-amplifying explosion events. We describe the mathematical formulation, implementation, and emergent patterns observed under various parameter regimes. The open-source release provides researchers with a reproducible testbed for studying hormetic dose–response, oscillatory threshold phenomena, and synthetic endocrine disruption scenarios.

Index Terms—generative regulatory automaton, hormonal explosion, endocrine dynamics, computational biology, agent-based simulation

I. INTRODUCTION

Rapid hormonal fluctuations—termed “hormonal explosions”—are fundamental to puberty, stress responses, and metabolic switching [1]. While differential equation models capture smooth endocrine rhythms, they often fail to reproduce the abrupt, switch-like surges observed *in vivo*. Recent advances in generative regulatory networks [2] offer a more expressive formalism, but their application to endocrine systems remains unexplored.

The **GRA-Hormonal Explosion of Subject** project fills this gap by constructing a subject-centric digital twin whose internal glandular layers are governed by a Generative Regulatory Automaton (GRA). The GRA evolves a hormone vector through discrete-time, rule-based interactions, enabling both stochastic baseline secretion and deterministic feedback explosions. The accompanying visualizer renders the subject’s state as a dynamic avatar, making hormonal crises intuitively observable.

II. MODEL DESCRIPTION

A. Generative Regulatory Automaton (GRA)

A GRA is a tuple $\mathcal{G} = (\mathbf{H}, \mathbf{R}, \mathbf{W}, \tau)$ where:

- $\mathbf{H} \in \mathbb{R}^n$ is the hormone concentration vector (e.g., cortisol, adrenaline, testosterone, estrogen).
- $\mathbf{R} \in \mathbb{R}^m$ represents receptor sensitivities.

- $\mathbf{W} \in \mathbb{R}^{n \times (n+m)}$ is a weight matrix encoding stimulatory/inhibitory interactions.
- τ is a threshold function that triggers discrete rule execution.

At each time step t , the state updates according to

$$\mathbf{H}_{t+1} = \mathbf{H}_t + \Delta t \cdot (\mathbf{W}_H \cdot \sigma(\mathbf{H}_t \oplus \mathbf{R}_t) + \boldsymbol{\xi}_t), \quad (1)$$

where $\sigma(\cdot)$ is a saturating nonlinearity (e.g., $\tanh$), $\oplus$ denotes concatenation, and $\boldsymbol{\xi}_t \sim \mathcal{N}(0, \Sigma)$ is Gaussian noise. An **explosion** is defined as a transient event in which $\|\mathbf{H}_t\|$ exceeds a pre-set burst threshold B for at least k consecutive steps.

B. Subject Layer and Hormonal Explosion Trigger

The subject is represented as a finite-state machine whose nodes are emotional/behavioural states (e.g., calm, agitated, explosive). Transitions depend on the dominant hormone:

$$S_{t+1} = f(S_t, \text{argmax}(\mathbf{H}_t)). \quad (2)$$

An external stressor or circadian trigger can inject a pulse $\mathbf{p}$ into a specific hormone channel, initiating a cascade. If the resulting positive feedback loop (through $\mathbf{W}$) drives the system above B , a *hormonal explosion of subject* occurs—the avatar visually erupts and the internal hormone levels oscillate chaotically before returning to baseline.

III. IMPLEMENTATION

The codebase (`GRA-Hormonal-Explosion-Of-Subject`) is written in Python with a Pygame visualization layer. Key modules:

- `gra_core.py`: Implements (1) with configurable gland networks.
- `subject_layer.py`: Manages the subject’s state machine and animation triggers.
- `explosion_detector.py`: Detects burst events and logs metrics (peak amplitude, duration, recovery time).
- `config.yaml`: Parameter file for $\mathbf{W}$, thresholds, and noise.

The simulation runs in real-time; users can perturb the subject by clicking to inject synthetic hormones or by scheduling stressors.

IV. RESULTS AND EMERGENT PHENOMENA

A. Explosion Signatures

Under default parameters, the system exhibits three regimes:

- 1) **Quiescent**: Low-amplitude wandering around homeostatic set-point.
- 2) **Pulsatile**: Periodic, predictable bursts resembling ultradian rhythms.
- 3) **Chaotic explosion**: Aperiodic, high-amplitude surges with long recovery tails, triggered by crossing a critical manifold in $\mathbf{W}$.

Fig. 1 shows a representative explosion event. The positive feedback between cortisol and adrenaline creates a rapid spike, followed by inhibitory compensation that overshoots into a temporary deficit.



Fig. 1. Hormone concentrations during a triggered explosion. Cortisol (solid) and adrenaline (dashed) exhibit co-bursting behaviour. The grey band marks the explosion interval.

B. Sensitivity to Network Topology

We tested randomly generated $\mathbf{W}$ matrices. Networks with strong reciprocal excitation between two glands (e.g., hypothalamus–pituitary–adrenal axis motifs) were significantly more explosion-prone (Mann–Whitney $p < 0.001$). Sparser, feedforward-like topologies required larger external pulses to trigger an event.

V. DISCUSSION

The GRA-HES framework reproduces hallmark features of hormonal explosions: threshold-dependent triggering, all-or-nothing responses, and slow recovery. Its generative nature allows exploration of “what-if” endocrine disruption scenarios without animal experimentation. The open-source release (<https://github.com/qqewq/>

GRA-Hormonal-Explosion-Of-Subject) invites the community to extend gland libraries, incorporate pharmacokinetic delays, or couple the subject to multi-agent social simulations.

Limitations include the abstract, dimensionless hormone units and lack of spatial compartmentalisation. Future work will integrate a hormone transport layer (diffusion between glands) and calibrate parameters against empirical cortisol awakening response data.

VI. CONCLUSION

We presented GRA-Hormonal Explosion of Subject, a flexible computational model of acute endocrine surges built on a generative regulatory automaton. The simulation successfully generates realistic explosion dynamics and serves as an interactive educational and research tool. The repository is publicly available and actively maintained.

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